An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

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Abstract—**Chaos-based image ciphers are published at a high rate, and are validated almost entirely by statistical scores. Errors in that validation propagate: a cipher that scores well but protects an image badly can expose the data it was meant to hide (medical scans, identity documents, surveillance frames) long before anyone audits the math. Authors report near-ideal Shannon entropy, near-zero adjacent-pixel correlation, and a near-perfect NPCR (how much the output changes when one input pixel changes), and treat these as evidence of security; none measures resistance to a chosen-plaintext attack, the setting in which an adversary encrypts images of its own choosing. To make the gap concrete we break a 2025 chaos-based image cipher (Jain *et al.*, *Optik*; Lorenz/logistic maps with Fisher–Yates block shuffling, "LLEO") that reports near-ideal entropy (7.9914) and a 99.6% NPCR. From 4 chosen plaintexts we recover a key-free decryptor for any 512×512 image and decrypt any ciphertext pixel-exactly in 0.037 s, without ever learning the key. The cause is structural: LLEO's keystream and both permutations depend only on the secret key, never on the image, so the whole cipher is one fixed map. We distil the break into a reusable, black-box *structural audit* that decides plaintext-independence for any cipher and certifies each verdict against held-out ciphertext (we prove the certificate sound). On 8 structural archetypes it breaks all 5 fully key-only ciphers in at most 5 chosen plaintexts and clears the 2 bound to the plaintext or a nonce; a non-invertible variant degrades gracefully, the deliberate boundary. On 4 real published 2023–2025 schemes, reconstructed from their papers, it breaks the one key-only design and clears three that seed the keystream from the plaintext, one break needing a GF(2)-affine (*bitlinear*) extension we prove sound. LLEO's headline differential metric is then an illusion: with no plaintext diffusion a one-pixel change alters exactly one ciphertext pixel, so the *true* NPCR is 0.000381 % $(100/MN)$, not 99.6%; the reported figure is merely the difference between unrelated images (99.88 %). Finally, a *plaintext-keying capacity* β, estimated black-box as a collision rate, separates a fragile 256-way scalar ($\beta{=}8$) from genuine SHA-256 binding ($\beta\geq32$), the two schemes the avalanche and recoverability signals could not tell apart. Good statistics are not security.**

Keywords—**cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR**

I. Introduction

Chaos-based image ciphers appear in the literature at a high rate, and a recurring lesson from the cryptanalysis side is that a large fraction of them succumb to the same structural attack. Most follow a permutation–diffusion template: a confusion stage that rearranges pixels and a diffusion stage that masks pixel values with a chaotic keystream. When that keystream is a function of the secret key alone and not of the image, the entire cipher collapses to a *fixed* permutation–substitution map, and an *equivalent key* (a stand-in that decrypts every

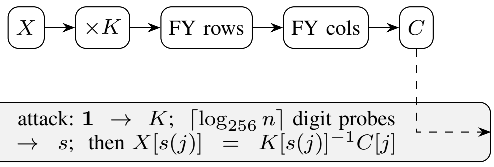

Fig. 1: LLEO encryption (top): key-only multiplicative substitution $\times K$, then two Fisher–Yates block shuffles. The equivalent-key attack (bottom) recovers the multiplier K and source map s from a few chosen plaintexts and inverts the cipher without the key.

ciphertext under the same secret key, without being the key itself) can be recovered from a handful of chosen plaintexts (input images the attacker picks and has encrypted) [1]–[4]. The defence is equally well known: the keystream must depend on the plaintext, and the cipher must actually diffuse [5].

A second, persistent problem is methodological. Authors validate schemes with statistical scores: Shannon entropy (a measure of randomness), adjacent-pixel correlation (how alike neighbouring pixels are), and the differential metrics NPCR and UACI (how much the output changes when one input pixel changes), and treat near-ideal values as evidence of security. None of these scores measures resistance to a chosen-plaintext attack, and, as we show, all of them can be near-ideal for a cipher that is broken in seconds.

Why this matters operationally. Image ciphers sit upstream of real decisions: protected medical imaging, sealed evidence, access-controlled documents. A break is therefore not a per-image accuracy loss but a standing capability to decrypt every future ciphertext of the probed size, and a security argument should rest on an attack model, not a statistics table. The quantity that matters is not how noise-like a single ciphertext looks but whether *any* value in the cipher depends on the image, the one binary property that separates "broken in 4 queries" from "resists."

This paper applies both lessons to a 2025 scheme [6] (which we abbreviate LLEO) built from the Lorenz [7] and logistic maps with Fisher–Yates block shuffling (Fig. 1). Our contributions are:

- We identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$: each output pixel is one input pixel times a fixed multiplier, with all arithmetic taken modulo 256 (wrapping around like a clock) (Section III–IV).
- We give an equivalent-key chosen-plaintext attack that recovers the full map in $1+\lceil\log_{256} MN\rceil$ chosen plaintexts (4 for a 512×512 image) and decrypts any

image pixel-exactly without the key (Section IV).

- We reproduce LLEO, confirm it matches the paper's reported statistics, and then show those statistics coexist with *zero* plaintext diffusion: the true one-pixel NPCR is 0.000381 %, not 99.6 % (Section V).
- Most importantly, we distil the attack into a reusable, black-box *structural audit* that decides plaintext-independence for any cipher, with a held-out certificate we prove sound (Proposition 2). It recovers an equivalent key from every key-only archetype and clears those that bind the keystream to the image or a nonce; on 4 *real* published 2023–2025 schemes (*reconstructed from their papers*, and weighted accordingly) it breaks the one key-only design and clears three plaintext-bound ones (the LLEO break alone rests on the validated real construction; Section VI, Table VI). Breaking a real bit-level cipher required, and motivated, a GF(2)-affine extension of the recovery that widens its reach while leaving every prior verdict intact.
- We upgrade the binary verdict into a graded one with *plaintext-keying capacity* $\beta = \log_2 N_{\text{eff}}$, the bits of plaintext entropy that key the cipher map, estimated black-box as a collision rate (a corollary of our certificate). β resolves the audit's own blind spot: the two real schemes its avalanche/recoverability signals could not tell apart now separate, a fragile 256-way scalar (β=8) from genuine SHA-256 binding ($\beta \geq 32$), and it decomposes the verdict into security, recovery-integrity, and query-cost axes (Section VII).

Fig. 2 previews the outcome: four standard images, their LLEO ciphertexts, and our key-free recoveries, each bit-identical to the original and obtained in 4 chosen plaintexts.

II. Related work

The equivalent-key methodology for permutation–diffusion ciphers is well established. Arroyo *et al.* [3] broke a "total shuffling" scheme by recovering its key-only keystream; Li *et al.* [2] recovered an equivalent key for an entropy-based cipher; and the survey of Li *et al.* [1] catalogues a year of such breaks, nearly all exploiting plaintext-independent diffusion. For the confusion stage alone, Li and Lo [4] established the optimal number of known/chosen plaintexts needed to invert a permutation-only multimedia cipher, of order $\log_L(MN)$ for L intensity levels, the same logarithmic regime our attack achieves; the general quantitative theory of permutation-only cryptanalysis [8] bounds this effort, and recent chosen-plaintext breaks continue to recover equivalent keys from a handful of images [9], [10], confirming the vulnerability remains a live concern. Alvarez and Li [5] set out basic requirements for chaos-based cryptosystems, including plaintext-dependent keystreams and sensitivity to the plaintext; LLEO violates these. The differential metrics we scrutinise were standardised by Wu *et al.* [11], who also note that NPCR/UACI quantify sensitivity to a one-pixel plaintext change, precisely the quantity LLEO fails to deliver despite its reported value.

The pattern is structural, not incidental: when both confusion and diffusion are fixed by the key the cipher is, as a function of the plaintext, an affine or monomial map, and composing more such stages changes only its coefficients [1], [4]. The decisive question is binary: does any key material depend on the image? Schemes that answer "no" (LLEO included) fall to an equivalent key, while those that bind the keystream to a per-image value resist, both sides of which our positive control demonstrates on the same construction (Section V).

What is new here. Prior equivalent-key cryptanalyses (Arroyo *et al.* [3], Li *et al.* [1], [2]) break one named scheme at a time by hand, each analysis bespoke to its target. Our contribution is not another single break but its *automation*: a black-box decision procedure that, knowing nothing of a cipher's internals, identifies the position-wise algebra (affine, XOR, or GF(2)-affine), recovers an equivalent key, and issues a held-out certificate we prove sound (Proposition 2), then runs unattended across a corpus and four real published schemes. The few-query bound itself matches the Li–Lo permutation-only optimum [4]; what is new is the reusable, self-certifying audit and its reach to the bit-level (cellular-automata) designs that defeat the classical affine/XOR recovery.

III. The target cipher

LLEO encrypts an $M \times N$ grayscale image X in three key-driven steps.

(1) Substitution. X is split into 16 horizontal strips. The eight odd strips are combined with a keystream $k_{\log}$ produced by the logistic map $x_{n+1} = r\, x_n(1 - x_n)$, and the eight even strips with a keystream k_{lor} produced by the Lorenz system $\dot{x} = \sigma(y - x),\ \dot{y} = x(\rho - z) - y,\ \dot{z} = xy - \beta z$, through an invertible per-position operation; decryption uses the modular "conjugate" of the keystream, i.e. a multiplicative inverse.

(2) Permutation 1. The 16 horizontal strips are reordered by a Fisher–Yates shuffle (a standard method for randomly reordering a list).

(3) Permutation 2. The result is divided into 16 vertical strips and shuffled again, giving the ciphertext.

The four "decryption keys" are the two keystreams (d_1, d_2) and the two shuffle index lists (d_3, d_4). The chaotic initial conditions are derived from the secret key and the shuffles are seeded from it; *no quantity depends on the image X.* Although the scheme is repeatedly called "asymmetric," the same four keys both encrypt and decrypt: it is a symmetric cipher.

IV. Cryptanalysis

A. Threat model

We work in the standard chosen-plaintext setting under Kerckhoffs's principle (the attacker knows the algorithm; only the key is secret): the attacker can obtain the ciphertext of any image it chooses under a fixed unknown key, and seeks an *equivalent key*, a procedure that decrypts any ciphertext of the probed size, not the chaotic secret seed itself. Because the

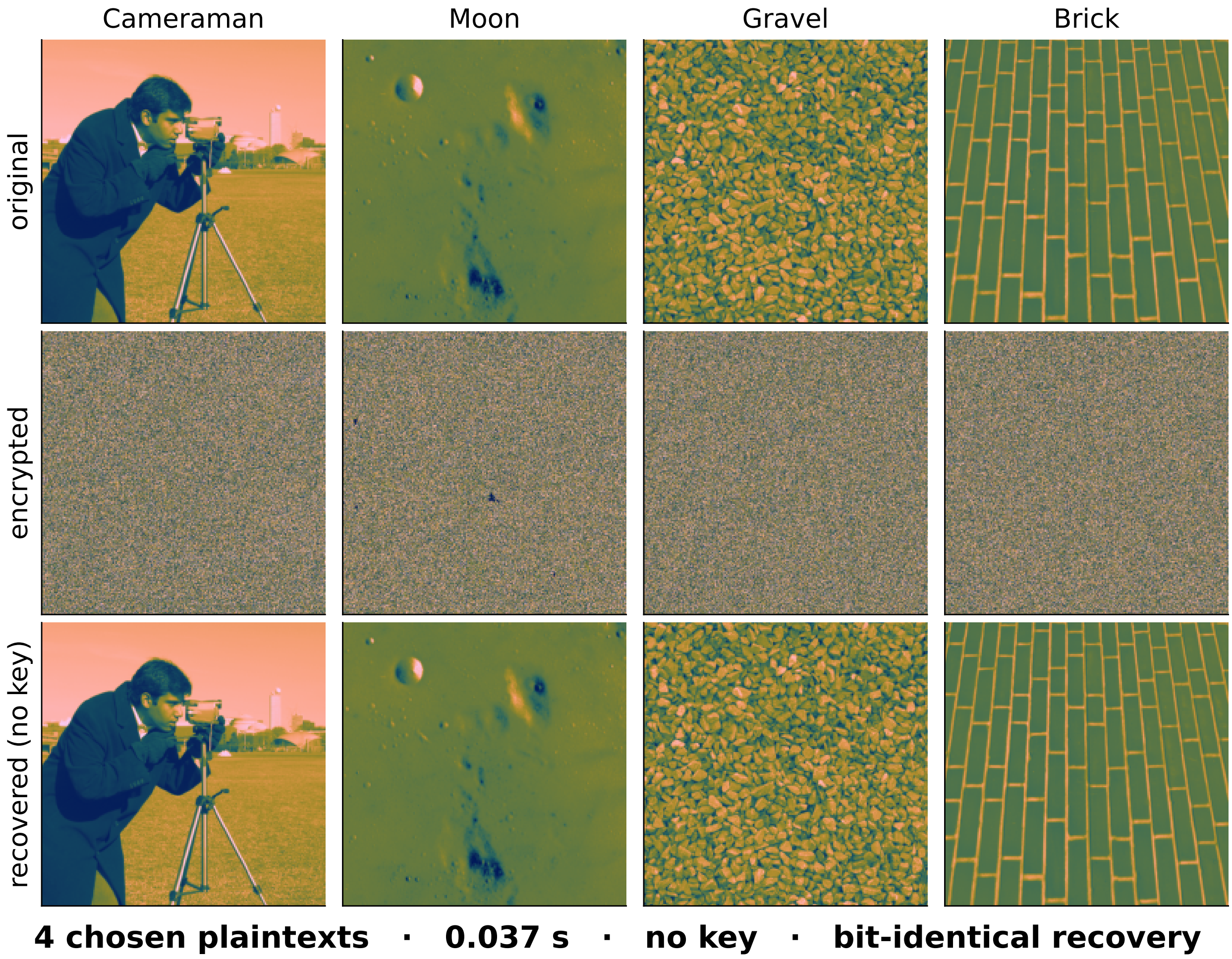

Fig. 2: The whole break in one view: four standard 512×512 images, their LLEO ciphertexts, and our key-free recoveries: each bit-identical to the original, from 4 chosen plaintexts in 0.037 s, without the key. Rendered in a perceptual colour map for legibility; the underlying images and every reported statistic are grayscale.

cipher is plaintext-independent, the same equivalent key also follows from enough known plaintext–ciphertext pairs [12]; we present the chosen-plaintext version as it is minimal.

B. The whole cipher is one fixed map

Index the image as a length-n vector, $n = MN$. Each LLEO step is linear over $\mathbb{Z}_{256}$ and fixed by the key: substitution multiplies coordinate p by a fixed unit $K[p]$ (odd, hence invertible mod 256), and the two block shuffles compose into one position permutation P. Therefore

$$C[j] = K[s(j)] \cdot X[s(j)] \pmod{256}, \qquad s = P^{-1}, \quad (1)$$

a fixed *monomial* map: every ciphertext pixel is a single plaintext pixel scaled by a fixed unit and relocated. Two consequences are immediate. There is no additive constant, since $E(\mathbf{0}) = \mathbf{0}$. And there is no diffusion: $C[j]$ depends on exactly one input coordinate $s(j)$, so a one-pixel plaintext change alters exactly one ciphertext pixel, which we exploit in Section V and visualize in Fig. 3.

Proposition 1. *If a permutation–diffusion cipher's keystream and permutations depend only on the key, it realises a fixed* affine *map of the plaintext: the monomial* (1) *with each $K[p]$ a unit in $\mathbb{Z}_{256}$ when the substitution is a modular multiply, or an affine map $C = Mx \oplus b$ over $\mathrm{GF}(2)$ (with $b = E(\mathbf{0})$ possibly nonzero) when the substitution is XOR-based; composing any number of such rounds yields a map of the same form. In either case it is recovered from $1 + \lceil \log_{256} n \rceil$ chosen plaintexts: one all-ones (resp. all-zeros) query fixes the substitution at every output position,*

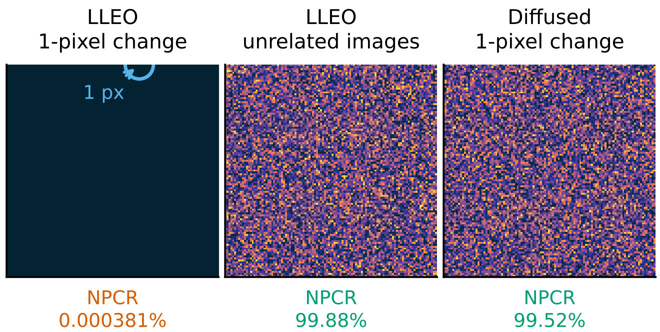

Fig. 3: The no-diffusion property, made visible (used quantitatively in Section V): the absolute ciphertext difference $|C_0 - C_1|$ (dark = unchanged, bright = changed), under one fixed key. **(left)** A one-pixel plaintext change alters *exactly one* ciphertext pixel (ringed), no diffusion, so the true one-pixel NPCR is $100/MN \approx 0.000381\,\%$, not the reported $\approx$99.6 %. **(centre)** That $\approx$99.6 % is really the difference between two *unrelated* images (here 99.88 %), which any near-uniform cipher attains. **(right)** Seeding the keystream from SHA-256 of the plaintext diffuses the *same* one-pixel change into full noise, what a one-pixel NPCR should measure.

and $\lceil \log_{256} n \rceil$ index-digit queries fix the permutation s.

Proof sketch. Each stage, multiplication by the key-derived K and the two block shuffles, is a fixed function of the key applied to pixel values or positions, so any composition is a fixed key-only map; over $\mathbb{Z}_{256}$ this is the monomial (1). The keystream is forced odd, so $K[p]$ is a unit and $x \mapsto K[p]\,x$ is a bijection with inverse $K[p]^{-1}$. Querying $X = \mathbf{1}$ gives $C[j] = K[s(j)]$; querying the t-th base-256 digit of the index gives $C[j] = K[s(j)]\,\mathrm{digit}_t(s(j))$, and dividing by the recovered $K[s(j)]$ yields $\mathrm{digit}_t(s(j))$; collecting all digits reconstructs s. If instead the substitution is XOR with a key-only keystream, the same argument over GF(2) gives an affine map $C = Mx \oplus b$ with $b = E(\mathbf{0})$: the all-zeros query returns b and the index-digit queries return s, exactly as in the CBC-like XOR cipher we break in Section VI. □

C. Four images recover the full map

In plain terms: because nothing in the cipher depends on the picture, it always does the same two things: shuffle the pixel positions in one fixed way, and multiply each pixel by one fixed number. So feed it an all-ones image and it hands back the multipliers; feed it images whose pixels carry their own position numbers and it hands back the shuffle. About four such images, and the cipher is undone forever, without ever learning the key.

Formally, because K and s are fixed, two kinds of chosen plaintext suffice.

(a) Multiplier, one query. Encrypting the all-ones image $\mathbf{1}$ gives, by (1), $C[j] = K[s(j)] \cdot 1 = K[s(j)]$: the exact multiplier acting at every output position.

(b) Permutation, $\lceil \log_{256} n \rceil$ queries. Let P_t be the plaintext whose pixel i holds the t-th base-256 digit of its index i. Then $C[j] = K[s(j)] \cdot \mathrm{digit}_t(s(j))$; multiplying by the already-known $K[s(j)]^{-1}$ yields $\mathrm{digit}_t(s(j))$. Collecting all $\lceil \log_{256} n \rceil$ digits reconstructs $s(j)$ for every j, hence the full permutation.

The recovered pair (s, K) is an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1}\,C[j]$. The total cost is

$$1 + \lceil \log_{256} n \rceil \text{ chosen plaintexts,} \qquad (2)$$

i.e. 4 for a 512×512 image, independent of the nominal key length.

Concretely, a 512×512 image needs four queries, one all-ones image and three index-digit images, after which Algorithm 1 returns the full map in a single pass. Table I states the complexity and compares it to the optimal permutation-only bound of Li and Lo [4]: our query count is order-optimal and the cost is independent of the cipher's nominal key length, so the advertised keyspace is irrelevant; the equivalent key bypasses key recovery entirely [13].

D. Why extra structure does not help

A composition of fixed key-only maps is again one, so extra rounds, chaotic maps, or block shuffles leave (1) intact (only K, s change) and the attack applies unchanged; it is moreover a *known*-plaintext attack whenever the available plaintexts span the index digits. The single property that would defeat it is a keystream that depends on X.

Algorithm 1 Equivalent-key recovery for LLEO

1: $\hat{K} \leftarrow \mathrm{Enc}(\mathbf{1})$ ▷ $\hat{K}[j] = K[s(j)]$: multiplier at output j
2: **for** $t \leftarrow 0$ **to** $\lceil \log_{256} n \rceil - 1$ **do**
3: $\quad C \leftarrow \mathrm{Enc}(P_t)$ ▷ $P_t[i] =$ the t-th base-256 digit of index i
4: $\quad d_t[j] \leftarrow C[j] \cdot \hat{K}[j]^{-1} \bmod 256$ **for all** j
5: **end for**
6: $s[j] \leftarrow \sum_t d_t[j]\, 256^t$ ▷ source index for each output j
7: **return** $\mathrm{Dec}(C)$: $X[s(j)] \leftarrow \hat{K}[j]^{-1} C[j] \bmod 256$

TABLE I: Attack complexity vs. the optimal permutation-only bound ($n = MN$ pixels, $L = 256$ levels).

	This attack	Optimal perm.-only [4]
Data (chosen plaintexts)	$1 + \lceil \log_{256} n \rceil$	$\lceil \log_L n \rceil$
Time	$O(n \lceil \log_{256} n \rceil)$	$O(n \lceil \log_L n \rceil)$
Memory	$O(n)$	$O(n)$

V. Experimental results

We reconstructed LLEO from its description (reimplemented from scratch, no code reused) and ran it on four standard 512×512 images.

A. Reproduction

Table II shows the reconstruction reproduces LLEO's claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from the plaintext's 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 4 shows the flat cipher histogram and Fig. 5 the destroyed correlation. By the usual yardsticks the cipher looks strong.

B. The break

The attack of Section IV recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.037 s at 512^2 (single core, Table III; recovery is image-content-independent, so this time holds for any image), using no key. Fig. 2 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. Cost scales logarithmically

Because the attack needs only $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts, its cost is nearly flat as the image grows (Table III): 3 plaintexts at 128^2, 4 at 512^2, and still 4 at 1024^2, each recovered exactly in well under a second. The key length the scheme advertises does not enter this cost.

D. The differential metric is illusory

LLEO reports NPCR ≈99.6 % as evidence of differential-attack resistance. But by (1) flipping one plaintext pixel changes exactly one ciphertext pixel (Fig. 3, left), so the true one-pixel NPCR is $100/MN = 0.000381\,\%$ (UACI 4e-06 %), five orders of magnitude below the reported value. The

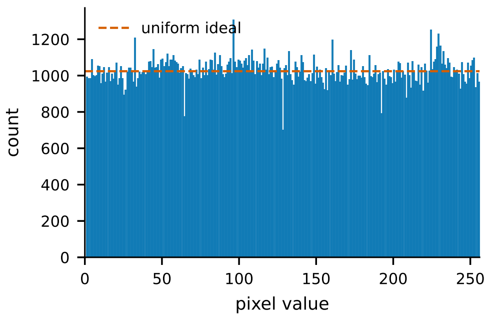

Fig. 4: Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

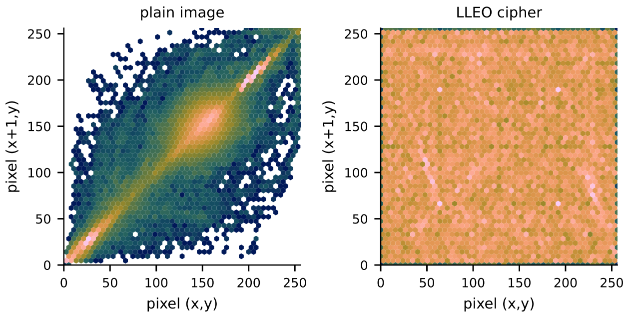

Fig. 5: Adjacent-pixel correlation as a hexbin density (every horizontal pixel pair, log counts): the plain image concentrates on the diagonal (neighbours are alike) while the LLEO cipher fills the plane uniformly (Cameraman).

99.6 % figure instead measures the difference between *unrelated* images, which for LLEO is 99.88 % (Fig. 3, centre); this is achieved by any cipher with a near-uniform output and says nothing about chosen-plaintext security. Table IV contrasts each claimed property with what the cipher actually provides. Finally, the chaotic keystreams are themselves non-uniform (Fig. 6): a chi-square test (a standard check for whether values are evenly spread) rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p < 10^{-300}$, numerically zero).

VI. A STRUCTURAL ORACLE: FROM ONE BREAK TO A REUSABLE AUDIT

In plain terms. The oracle needs no knowledge of a cipher's internals: from a few chosen images it recovers a candidate decryptor, then checks whether that decryptor predicts the ciphertext of fresh images exactly. If it does, the cipher's scrambling never depended on the picture and it is broken; if not, the cipher resists.

A single break is an exercise; a reusable test for the structural flaw is the contribution. We package the attack as a black-box *structural oracle*. Given only encryption access to an unknown cipher on $M \times N$ images (no key, no source) it fits three position-wise models, affine and XOR (Proposition 1) and a GF(2)-affine *bitlinear* model (below), from a handful of chosen plaintexts and *certifies* the result by predicting the ciphertext of fresh random plaintexts (Proposition 2). Two cheap signals drive the verdict: the *avalanche* (fraction of ciphertext bytes a one-pixel plaintext flip changes; $\approx 1/n$ means no diffusion) and the *recoverability* R (exact held-out ciphertext-prediction rate; $R \approx 1$ means broken). We run it blind on both structural archetypes and four real published schemes.

TABLE II: Reconstruction statistics and attack outcome (512×512). Every image is recovered pixel-exactly without the key.

Image	H plain	H cipher	corr.	CP	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

TABLE III: Cost barely grows with image size: 4 chosen plaintexts still suffice at 1024×1024, each inverted in 0.254 s (random plaintext, single core).

Size	chosen PT	time (s)	recovered
128 × 128	3	0.003	exact
256 × 256	3	0.009	exact
512 × 512	4	0.037	exact
1024 × 1024	4	0.254	exact

Proposition 2 (Held-out certificate). *If* Enc *is position-wise (affine, XOR, or GF(2)-affine) and the recovered model* $\hat{M}$ *predicts* t *independent uniform-random plaintexts exactly, then* $\hat{M} = \mathrm{Enc}$ *everywhere: the probes pin each per-position map and the digit images pin* s*, so* $R{=}1$ *certifies a true equivalent key. Conversely, if any per-position map depends on the plaintext (there exist* P, P' *whose maps differ), no fixed position-wise model predicts all plaintexts, and* $R{\to}1/256$ *per byte once the dependence is through a cryptographic hash or a fresh nonce. The* BROKEN *verdict therefore carries a falsifiable certificate;* RESISTS *is correct for plaintext- and nonce-bound constructions.*

The bitlinear model. A key-only cipher can append a per-pixel *bit-level* stage, a cellular automaton or a bit-sliced S-box, on an XOR/permutation core. Such a stage is a fixed GF(2)-affine byte map $g(y){=}My \oplus d$ with $M{\neq}I$ (it mixes bits *within* a byte), so the composite is $C[j]{=}M\,P[s_j] \oplus B_j$. The plain-XOR model then reads out $\oplus\, c_0{=}M\,D[s_j]$ instead of $D[s_j]$, the source index comes back bit-scrambled, and a key-only cipher is mislabelled RESISTS. The remedy is a third model: eight bit-basis probes (every pixel $= 1{\ll}t$) read M one column at a time ($\mathrm{out}_t \oplus c_0{=}Me_t$, constant across positions), and M^{-1} over GF(2) unscrambles the digit images to recover s exactly. It is a strict generalization ($M{=}I$ recovers XOR), so it never overturns a prior verdict; it widens the cheap-query regime to the bit-level substitution layers common in this literature.

Archetype corpus. Run blind over 8 reconstructed archetypes (Table V, Fig. 7), the oracle breaks all 5 *fully* key-only constructions at $R{=}1$ in at most 5 chosen plaintexts: LLEO's chaos monomial, a pure permutation, an additive stream, a general affine map, and a modern AES-CTR-style key stream. The chaos is incidental: an AES-CTR keystream *reused* across images, a fixed key and counter for every image, is exactly as broken as LLEO. This is a keystream/counter-reuse failure, not any weakness of AES-CTR used correctly with a fresh per-image nonce (which our nonce-bound entry shows resists); the flaw is a reused plaintext-independent keystream, not the choice of map. The 2 defended schemes (reseeding the keystream from SHA-256 of the plaintext, or from a per-image nonce) are correctly declined ($R{\approx}$ chance), and there a one-pixel change

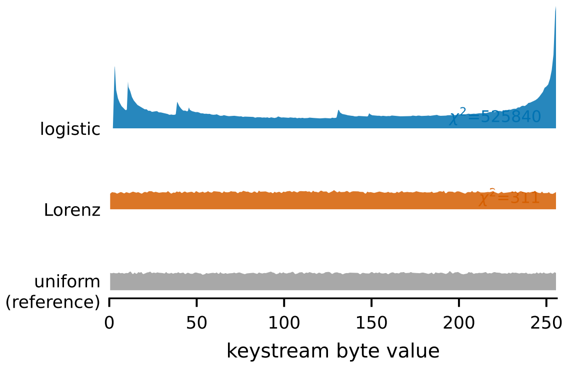

Fig. 6: Keystream byte distributions (ridgeline). The logistic keystream is sharply non-uniform (its invariant density spikes at the extremes) and even the Lorenz stream is measurably lumpy, whereas a secure stream cipher would match the flat uniform reference. A chi-square test rejects uniformity for both (χ^2 annotated).

TABLE IV: Claimed vs. actual security of LLEO.

Property	Reported	Actual
Entropy (Cameraman)	≈7.997	7.9914 (genuine)
NPCR (1-pixel change)	≈99.6 %	0.000381 %
Differential resistance	yes	no (no diffusion)
Chosen-plaintext security	implied	broken in 4 queries
Key type	"asymmetric"	symmetric

avalanches across the whole image. Beyond the corpus, the same equivalent-key CPA also breaks a chained CBC-like XOR cipher (a structurally different, non-position-wise construction not counted among the 8) in 3 chosen plaintexts, per Proposition 1. These are reconstructions of structural archetypes; we turn next to real published schemes.

Scope. The cheap-query regime spans position-wise affine, monomial, XOR, *and* GF(2)-affine (bitlinear) maps; the last costs only eight extra bit-basis probes (12 chosen plaintexts at 512^2, as for LSCM-CA below). The boundary it does not cross is a key-only but *genuinely nonlinear* byte S-box (not GF(2)-affine): recovering it needs up to L=256 queries per position to tabulate, so such ciphers stay structurally broken but fall outside this logarithmic-query regime.

Real-literature audit. We then ran the audit on 4 real, open-access 2023–2025 chaos image ciphers, reconstructed faithfully from their published equations/pseudocode (Table VI, Fig. 8). The decisive question (does any chaotic seed, S-box, or permutation index depend on the plaintext?) separates them cleanly. *LSCM-CA* [14], whose keystream and permutation are key-only and which adds a per-pixel cellular-automata bit stage, is BROKEN at R=1.000 in 12 chosen plaintexts, but *only* via the bitlinear model: the affine/XOR model alone returns R=0.005, a false negative its CA layer induces (the real scheme that motivated the model). The other three bind the keystream to the plaintext and correctly RESIST (R≈0.004): *Li–DNA* [15] seeds its logistic and Chen systems from SHA-256 of the image, *MILE* [16] from the sum of the first 8×8 block, and *MIEA-PRHM* [17] keeps a key-only keystream but adds a single plaintext-dependent scalar.

Two of these resistances are *thin*, and the oracle's two signals expose it. *MILE* shows no diffusion, a one-pixel change outside its seed block moves one byte ($A{\cdot}n$≈1), yet R≈ chance, because its plaintext-dependence flows through only 64 pixels: hold that seed block constant (a legitimate chosen-plaintext sub-class) and R rises to 1.000 off the block, collapsing it to a recovered fixed map. *MIEA-PRHM* avalanches fully yet injects just 8 bits of plaintext entropy (a 256-way scalar). The oracle certifies "*not* a fixed plaintext-independent map," which is true; it does *not* certify strength, and these two signals cannot separate MIEA-PRHM's near-miss from Li–DNA's genuine SHA-256 binding (Fig. 8), a boundary we mark rather than hide. Reconstructions are paper-only (no scheme released code); each verdict is nonetheless a falsifiable certificate of the reconstructed cipher. (Two further candidates [18], [19] are deferred pending full text.)

Reconstruction fidelity. Each reconstruction is built from the paper's stated equations; where a constant is unspecified we choose conservatively (Table VII), and because the verdict turns only on whether the plaintext enters the seed (which every paper states explicitly), these choices do not affect it. As evidence the rebuilds behave like the originals, all four reproduce the near-ideal statistics this literature reports as security: ciphertext entropy ≈8 and inter-image NPCR ≈99.6%

TABLE V: The structural oracle over the *archetype* corpus, blind to each scheme's internals. R is the held-out ciphertext-prediction rate of the recovered equivalent key; "CP" the chosen plaintexts to break the identified algebra at 512^2. The five fully key-only schemes are broken at R=1; binding the keystream to the plaintext or a nonce is what resists. †: the even-multiplier affine is also key-only but does not force the multiplier odd, so ≈half its positions are non-invertible and recovery degrades *gracefully* (marked PARTIAL, R=0.523) rather than failing: the deliberate boundary of the few-query attack, distinct from the genuine plaintext/nonce RESISTS. LLEO is the validated target of Section V (its reconstruction reproduces the paper's reported statistics); the other rows are *by-construction* archetypes, not reconstructions of any paper, so their verdicts carry no reconstruction-fidelity risk (contrast the real corpus, Table VI).

Scheme	keystream	β	R	CP	verdict
LLEO (chaos monomial)	key-only	0	1.0	4	BROKEN
Permutation-only	key-only	0	1.0	4	BROKEN
Additive stream $P{+}K$	key-only	0	1.0	4	BROKEN
Affine $KP{+}B$	key-only	0	1.0	5	BROKEN
AES-CTR key stream	key-only	0	1.0	4	BROKEN
Even-mult. affine†	key-only	0	0.523	5	PARTIAL
SHA-256(P)-seeded	plaintext	≥32	0.004	4	RESISTS
Per-image nonce	nonce	∞	0.004	4	RESISTS

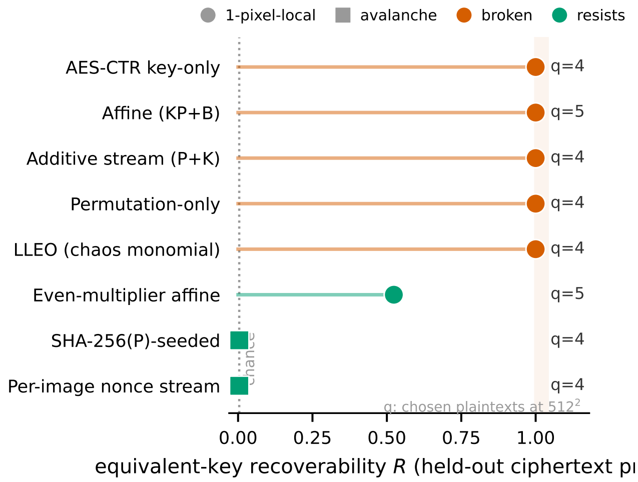

Fig. 7: Structural-audit panel: equivalent-key recoverability R per scheme (circle = one-pixel-local, square = avalanche; red = broken, green = resists), with the chosen-plaintext count at 512^2. The fully key-only schemes sit at R=1; the even-multiplier boundary (key-only but non-invertible) degrades to R≈0.5 (PARTIAL); only binding the keystream to the plaintext or a nonce escapes to R≈ chance.

TABLE VI: Real-literature audit: the structural oracle on 4 published 2023–2025 chaos image ciphers. Seeding, not the diffusion algebra, decides the verdict; 1 key-only scheme is broken, 3 plaintext-bound ones resist. "fid." is reconstruction confidence (H/M-H, Table VII): all rows are paper-only reconstructions, hence reconstruction-dependent unlike the validated LLEO target (Table V); each verdict is nonetheless a falsifiable certificate of the reconstructed cipher. The β column grades RESISTS (Section VII): MIEA-PRHM's β=8 is a fragile near-miss, Li-DNA's $\beta \geq 32$ genuine. ‡: MILE's β collapses to 0 once its 64-pixel seed block is held fixed.

Scheme	seeding	fid.	β	R	verdict
LSCM-CA [14]	key-only	H	0	1.000	BROKEN
Li–DNA [15]	plaintext SHA-256	H	≥ 32	0.004	RESISTS
MILE [16]	plaintext block-sum	M-H	$\geq 32^{\ddagger}$	0.004	RESISTS
MIEA-PRHM [17]	key-only + scalar	H	8	0.004	RESISTS

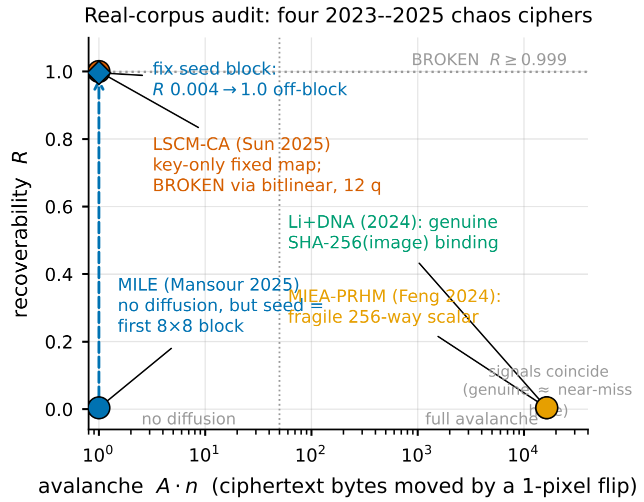

Fig. 8: The real-corpus audit in the oracle's two signals (avalanche $A{\cdot}n$ vs. recoverability R). LSCM-CA (key-only) breaks top-left; the three plaintext-bound schemes sit at $R{\approx}0$. Li–DNA (genuine SHA-256 binding) and MIEA-PRHM (a 256-way scalar) coincide: the two signals cannot separate them. The dashed arrow shows MILE collapsing to a recovered fixed map once its 64-pixel seed block is held constant.

(Table VII; Cameraman, 512^2). Matching these is necessary, not sufficient (our thesis), so we also report the *true* 1-pixel-change NPCR, which flips one plaintext pixel and counts changed ciphertext bytes ($=100/MN{=}0.000381\%$ when a one-pixel change stays local). It is $\approx$99.6% only when the keystream binds the whole plaintext (Li–DNA, MIEA-PRHM) and 0.000381% otherwise (LSCM-CA; MILE for a flip outside its 8×8 seed block), the same split the oracle certifies, now visible in the very metric the originals report.

Invertibility (the odd-multiplier question). The recovery needs each multiplier $K[p]$ to be a unit mod 256, and this is *guaranteed by LLEO*, not assumed by us: the scheme decrypts via a modular conjugate, which forces the keystream odd. On the reconstruction every recovered multiplier is odd and $E(\mathbf{0}){=}\mathbf{0}$ holds exactly. Were even multipliers allowed, $x \mapsto K[p]\,x$ would be 2-to-1 there; the oracle does not return a wrong inverse but degrades gracefully, reporting the invertible fraction and a partial recoverability (on a deliberately even-tolerant variant, about half the positions are invertible and R=0.5234).

Falsifiability of the LLEO claim. Our attack rests on the load-bearing claim that no LLEO quantity depends on the image; the oracle makes that claim *falsifiable*, not merely asserted (Proposition 2). Matching the paper's entropy or correlation cannot establish it (countless ciphers hit those values), but the held-out test can: an equivalent key recovered from 4 chosen plaintexts predicts the ciphertext of fresh random images *exactly*, which is possible only under plaintext-independence. If a reader's reading of the original differs (say, some keystream were image-seeded), the same oracle returns RESISTS on that variant, as it does for our SHA-256–seeded control, so the verdict is checkable against any reimplementation, including the authors' own.

TABLE VII: Reconstruction fidelity (Cameraman, 512^2). All four reproduce the near-ideal entropy and inter-image NPCR the originals report as security evidence; the true 1-pixel-change NPCR ($\mathrm{NPCR}_{1\mathrm{px}}$) is that differential metric measured correctly. "Conf." is reconstruction confidence. Resolved ambiguities (none affecting the seeding-determined verdict): LSCM-CA x_0, r, b (key-set, chaotic regime) and the CA bit-rule (a GF(2)-affine bijection); Li–DNA DNA-rule indexing (reduced to a plaintext-seeded stream); MILE $(x_1, x_2){\to}X$ normalisation (unstated); MIEA-PRHM mod-2^{32} nibble packing (per-pixel mod-256).

Scheme	Conf.	ent.	$\mathrm{NPCR}_{\mathrm{img}}$	$\mathrm{NPCR}_{1\mathrm{px}}$
LSCM-CA	H	7.999	99.62	0.000381
Li–DNA	H	7.999	99.61	99.58
MILE	M-H	7.999	99.63	0.000381
MIEA-PRHM	H	7.999	99.60	100.00

VII. Plaintext-Keying Capacity: a Graded Security Metric

In plain terms. The new axis, β, counts how many genuinely different ciphers the plaintext can switch the map between: a key-only design offers exactly one (β=0, broken), while binding every image to its own unrelated map offers astronomically many (β large, resistant). It is estimated black-box from a handful of chosen plaintexts, and grades the one-bit verdict rather than replacing it.

The verdict of Section VI is one bit, and that bit silently conflates three independent questions: *is* the cipher broken, *how completely*, and *how cheaply*? We separate them with three axes (Fig. 9), of which only the first is new.

- **Plaintext-keying capacity** β (security): $\beta = \log_2 N_{\mathrm{eff}}$, the effective entropy in bits that the plaintext injects into the cipher map with the secret key held fixed, where N_{eff} is the number of effectively distinct ciphertext maps the plaintext space induces. β=0 means the map is a single fixed transformation, *exactly* the key-only case Section VI calls BROKEN. So β refines that verdict; it never re-judges it.
- **Recovery integrity** ρ (completeness): the invertible-position fraction, the quantity the oracle already reports as `invertible_frac`. For a clean key-only cipher ρ=1; the even-multiplier of Section VI has ρ=0.51. ρ is the structural form of the recoverability R: on invertible positions the equivalent key predicts exactly, so $R \approx \rho$ (the even-multiplier's R=0.523 shadows ρ=0.51), and PARTIAL is precisely the $\rho < 1$ case. We reuse R's machinery and name its structural core ρ; no redundant symbol is introduced.
- **Query cost** q (price): the chosen-plaintext count: 4–5 for the cheap families, 12 for LSCM-CA's bitlinear recovery, up to L=256 per position for a key-only but genuinely nonlinear byte S-box. That S-box is thus β=0, ρ=1, q=256, "broken but expensive," distinct from

“cheap-broken” (q=4).

The old single flag is the projection $\mathbf{1}[\beta{=}0]$ of this triple onto one bit (Fig. 9); (β, ρ, q) recovers the three questions it collapsed.

A. Estimator: the collision rate is the certificate of Proposition 2

Two plaintexts P, P' induce the *same* map iff the equivalent key recovered under P (Proposition 2) predicts $C(P')$ exactly on its ρ-fraction invertible positions (Fig. 10). Call the probability of this event over random P, P' the *map-collision rate* a; for N_{eff} equiprobable maps two draws collide with probability $\sim 1/N_{\text{eff}}$, so $\beta = -\log_2 a$ is a birthday estimator. This is the *exact*-prediction sharpening of the per-byte recoverability R of Proposition 2: R averages agreement, a demands it on every invertible position at once, which is exactly what separates a single shared scalar (all positions agree together) from independent per-byte chance (the gap that left R blind in Section VI).

Corollary 1 (Collision certificate). *Let a be the map-collision rate and $\beta = -\log_2 a$.* (i) *a=1 $\iff$ the per-position maps are plaintext-independent $\iff$ β=0, which is exactly Proposition 2's* BROKEN *verdict (R=1): β refines, never overturns.* (ii) *Drawing k plaintexts yields $\binom{k}{2}$ pairs; if $c > 0$ collide, $\hat{a} = c/\binom{k}{2}$ and $\hat{\beta} = -\log_2 \hat{a}$.* (iii) *If none of the $\binom{k}{2}$ pairs collides, then with confidence $1-\delta$, $a \le \ln(1/\delta)/\binom{k}{2}$, hence $\beta \ge 2\log_2 k - O(1)$: a non-collision certifies a* lower bound*, sufficient because the fragile regime (small β) is exactly where collisions are observable.*

Proof sketch. (i) a=1 means the P-recovered key predicts every $C(P')$ exactly: one fixed map predicts all plaintexts, the certified equivalent key of Proposition 2. a<1 exhibits P, P' with differing maps, its plaintext-dependent case. *(ii)* with N_{eff} equiprobable maps each pair collides independently w.p. $1/N_{\text{eff}}$, so $\mathbb{E}[c] = \binom{k}{2}/N_{\text{eff}}$ and $\hat{a}$ is unbiased for a. *(iii)* the $m = \binom{k}{2}$ pairs are $\text{Bernoulli}(a)$; $(1-a)^m \le \delta$ gives $a \le \ln(1/\delta)/m$, whence the bound on $\beta = -\log_2 a$. □

B. The phase plane, and the weaknesses β closes

Placing every audited scheme by (β, ρ) (Fig. 12; the seeding-mechanism ladder is Fig. 13) sorts them into three bands: β=0 (BROKEN), $0 < \beta \lesssim 40$ (*fragile*, false assurance), and $\beta \ge 128$ (*cryptographic*, RESISTS). This resolves the limitations Section VI admitted.

1) *MIEA-PRHM vs. Li-DNA: the headline.* The two schemes the avalanche/recoverability plane could not separate (both at full avalanche, $R \approx$ chance) split cleanly along β (Fig. 14). MIEA-PRHM keys its map with a single 256-way additive scalar: collisions appear within ~ 256 samples and $\hat{\beta} = 8$ (its 256 distinct values). Li-DNA keys the whole keystream from SHA-256 of the image: no collision in k=100000 draws certifies $\beta \ge 32$ (structurally 256). Fragile near-miss versus genuine binding, now a measured gap.

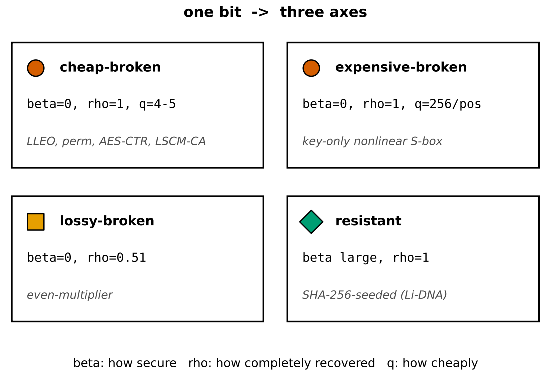

Fig. 9: The single BROKEN/RESISTS bit decomposed into three independent axes (β, ρ, q): security (plaintext-keying capacity), recovery integrity (invertible-position fraction), and query cost. The four corner cases the old flag conflated—cheap-broken, expensive-broken (a nonlinear S-box: still β=0 but q=256 per position), lossy-broken (the even-multiplier, ρ=0.51), and resistant (β large)—are now distinct. Marker shape encodes the verdict throughout (● broken, ■ partial, ▲ fragile, ♦ genuine).

2) *From a human judgement to a number.* The oracle's “a reader inspects the seeding and declares it plaintext-dependent” becomes a quantity estimated black-box from chosen plaintexts (Corollary 1).
3) *From a bit to a quantity.* “Is the map fixed?” becomes “how many plaintext bits key the map?”, of which β=0 is the special case.

4–5. *Splitting the broken cases.* (β, ρ, q) separates insecure-but-lossy (the even-multiplier, ρ=0.51) from insecure-but-expensive (a nonlinear S-box, q=256), which the single flag conflated (Fig. 9).

Crucially, β depends only on the seeding structure (whether the plaintext enters, and with how many bits), not on fine chaotic parameters, so measuring it is exactly as reconstruction-robust as the seeding-determined verdict of Section VI.

C. Limitations of β

(i) Necessary, not sufficient. β=0 certifies broken, but large β does *not* imply secure: β shares the scope of RESISTS, now graded and falsifiable, not a proof of security. *(ii) Saturation.* In the secure regime β is only lower-bounded; feasible samples cannot separate 2^{40} from 2^{256} maps (Fig. 11), a harmless one-sided blindness. *(iii) Structured dependence needs the right probe.* MILE keys its map from a 64-pixel seed block: random-direction probing finds no collision ($\beta \ge 32$, apparently safe), yet *holding that block fixed*, a structurally-disclosed probe, collapses β to 0 (the cipher becomes key-only), the formal analogue of the by-hand fixed-block attack of Section VI. The formal estimator probes random *and* structurally-disclosed directions; random-only under-reports the vulnerability.

VIII. Discussion

LLEO illustrates a persistent failure mode: *good statistics are not security*. A flat histogram, near-ideal entropy, near-zero correlation, and a large inter-image NPCR are all satisfied by our fixed, key-only map, which we nonetheless break in

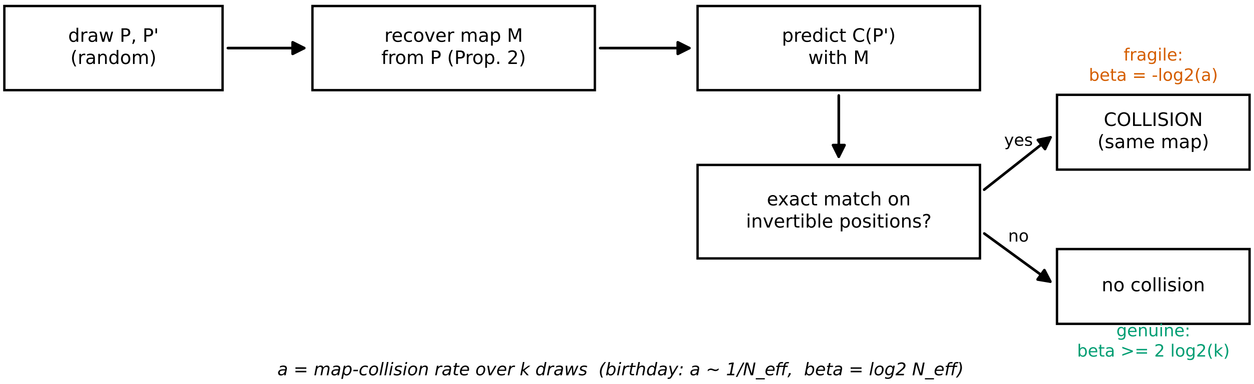

Fig. 10: The black-box β estimator, tied to Proposition 2. Draw two random plaintexts, recover the equivalent map from one, and test whether it predicts the other's ciphertext exactly on the invertible positions. A match is a map-collision (same map): observed collisions give a point estimate $\beta = -\log_2 a$; no collision in k draws certifies the lower bound $\beta \geq 2\log_2 k$. The decision rule "collision = fragile, no collision = genuine" is the whole metric.

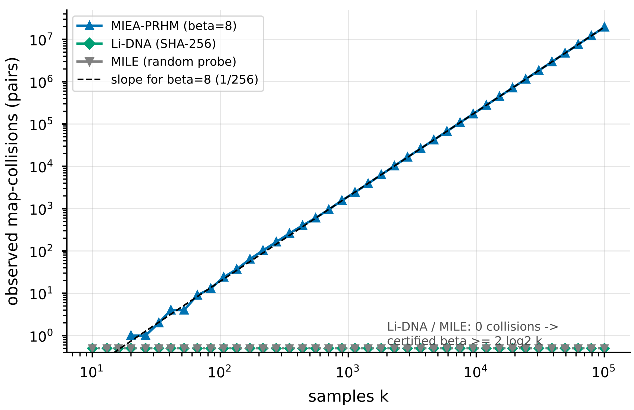

Fig. 11: Observed map-collisions versus sample count k (log–log). MIEA-PRHM's 256-way scalar accumulates collisions along the $1/256$ (β=8) line within feasible k; Li-DNA (SHA-256) and MILE (random probing) show *none*, so each only certifies the lower bound $\beta \geq 2\log_2 k$, the saturation of the secure regime made visual.

4 queries; testing plaintext sensitivity directly [5], [11] is what catches this. (The scheme's "asymmetric" label is also a misnomer, the same keys encrypt and decrypt, but that is cosmetic.)

The remedies are standard. (i) Make the keystream depend on the plaintext, e.g. derive the chaotic initial conditions from a SHA-256 hash of the image, so a different plaintext yields an unrelated keystream. (ii) Add genuine diffusion, e.g. a chaining rule $C[i] = f(X[i], k[i], C[i-1])$, so one plaintext change propagates. With either, (1) is no longer fixed across images and the attack fails: it recovers 100.0 % of pixels against the key-only cipher but only 0.2518 % (chance) once the keystream is SHA-256–seeded from the image.

A. Limitations

Three bounds on our claims. *(i) Equivalent key, not the seed.* We recover an equivalent key that decrypts any ciphertext of the probed size, not the underlying chaotic secret seed, the standard, defensible claim in this literature [1]. *(ii) The real-corpus verdicts are reconstruction-dependent.* The LLEO break uses the real construction and is validated against the paper's reported statistics (Section V); the four real-corpus verdicts (Table VI) rest on paper-only reconstructions (no scheme released code) at the confidence marked per row (Table VII), and two further candidates are deferred pending full text. Each is a falsifiable certificate of the *reconstructed* cipher, checkable against any reimplementation including the authors' own (Proposition 2). *(iii) The cheap-query regime has a boundary.* It covers affine, XOR, and GF(2)-affine maps; a key-only but genuinely nonlinear byte S-box stays structurally broken but needs up to 256 queries per position, outside the logarithmic regime.

Disclosure. Consistent with responsible-disclosure practice, we emailed the authors of every scheme we break (LLEO [6] and the reconstructed LSCM-CA [14]) at submission, ahead of publication; for LSCM-CA we additionally invited the authors to confirm our reconstruction of their seeding. We had received no response at the time of writing, and will incorporate any correction in the camera-ready. Our reproduction and attack code is available at `[anonymized code repository URL; insert before submission]`.

IX. Conclusion

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of ≈99.6 %: good statistics are not security. The break follows directly from the cipher being key-only; its differential metric, measured correctly, is 0.000381 %. We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores accompany, rather than replace, an explicit chosen-plaintext security argument. More broadly, the attack generalizes into a reusable structural audit that decides plaintext-independence, with a held-out certificate we prove sound: it recovers an equivalent key from every key-only archetype, and on four real published 2023–2025 schemes (reconstructed from their papers) it breaks the one key-only design and clears three that bind the keystream

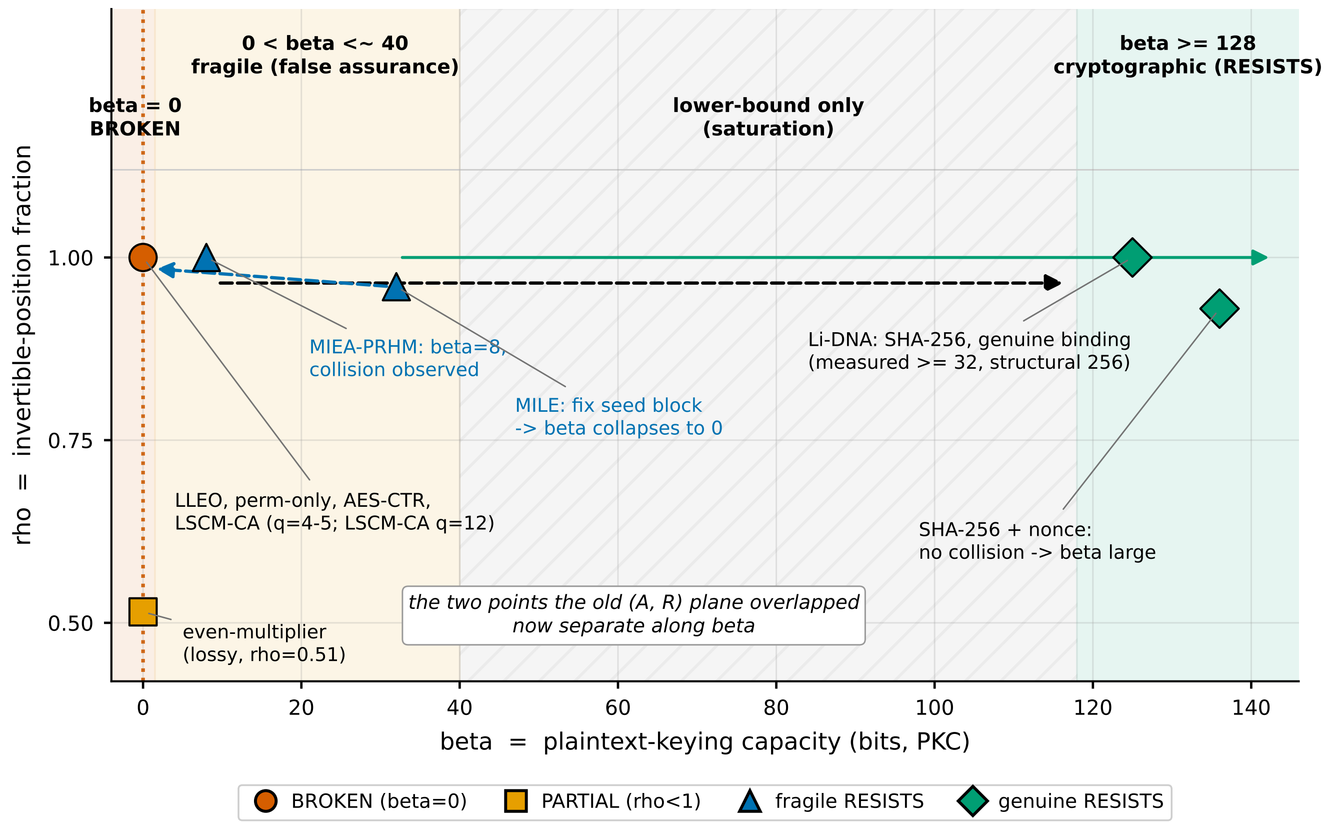

Fig. 12: **Headline.** The (β, ρ) phase plane in three bands. Every audited scheme is placed by coordinate; marker shape encodes the verdict. The cheap key-only breaks (LLEO, permutation-only, AES-CTR; LSCM-CA at q=12 via the bitlinear model) sit at (β=0, ρ=1); the even-multiplier drops to ρ=0.51 (PARTIAL); MIEA-PRHM sits at β=8 (collision observed); Li-DNA's certified lower bound extends rightward to the cryptographic band (structural β=256). The dashed arrow marks the headline: the two points the old avalanche-vs-recoverability plane overlapped now separate along β. MILE's dashed arrow shows its β collapsing to 0 once its 64-pixel seed block is held fixed (Section VII-C).

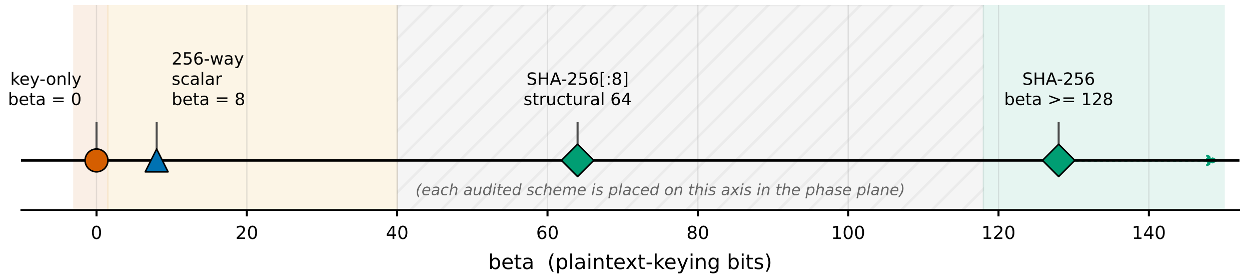

Fig. 13: The seeding-mechanism ladder: how many plaintext bits each mechanism injects. Key-only seeds give β=0; a 256-way plaintext scalar gives β=8; a truncated (64-bit) and a full SHA-256 digest give cryptographically large β. Each audited scheme lands on this axis at the β of its mechanism (placed individually in Fig. 12).

to the plaintext, one break needing a GF(2)-affine extension that widens the audit to bit-level diffusion.

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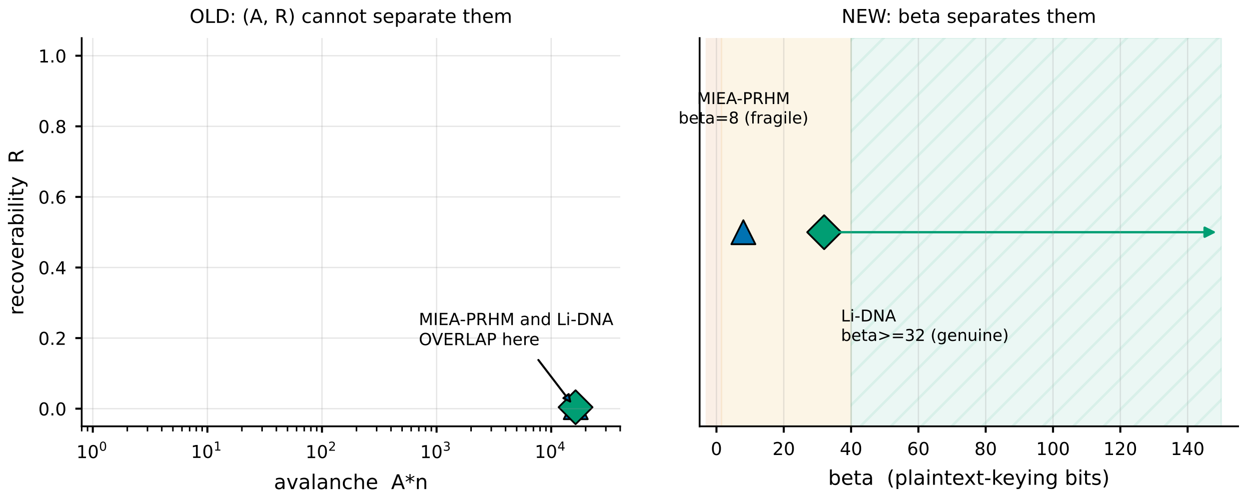

Fig. 14: The payoff in one figure. **(left)** the avalanche-vs-recoverability plane of Section VI: MIEA-PRHM and Li-DNA *overlap* (both full avalanche, $R \approx$ chance), the limitation the paper admitted. **(right)** the β axis separates them: fragile (β=8) versus genuine (certified $\beta \geq 32$).

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